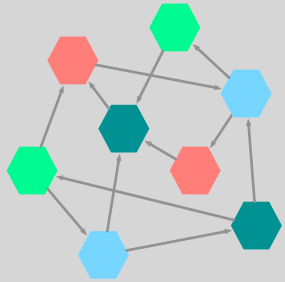
initial (annotated) scene graphs
(sparse, arbitrary)

A beside B
C over D

convert to RDF
and
insert into KG



knowledge graph
hosting NeSy4VRD
OWL ontology
VRD-World

extract from KG
and
convert to NeSy4VRD

logically augmented scene graphs
(denser, more consistent)

active mix of link inference semantics:
symmetry, **equivalence**

A beside B
C over D
B beside A
A nextTo B
B nextTo A

active mix of link inference semantics:
symmetry, **inverses**, **equivalence**

A beside B
C over D
B beside A
D under C
A nextTo B
D beneath C
B nextTo A

relevant **VRD-World** declarations (OWL **axioms**):

vrd:beside rdf:type **owl:SymmetricProperty**
vrd:nextTo rdf:type **owl:SymmetricProperty**
vrd:nextTo **owl:equivalentProperty** vrd:beside
vrd:under **owl:inverseOf** vrd:over
vrd:beneath **owl:equivalentProperty** vrd:under